



US 20250286750A1

(19) **United States**

(12) **Patent Application Publication**
ZHAN

(10) **Pub. No.: US 2025/0286750 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **COMMUNICATION INTERACTION
METHOD, APPARATUS, DEVICE, AND
STORAGE MEDIUM**

Publication Classification

(51) **Int. Cl.**
H04L 12/40 (2006.01)
G07C 5/08 (2006.01)
(52) **U.S. Cl.**
CPC **H04L 12/4015** (2013.01); **G07C 5/0808**
(2013.01); **H04L 2012/40273** (2013.01)

(71) Applicant: **Autel Intelligent Technology Corp.,
Ltd., Shenzhen (CN)**

(72) Inventor: **Jinyong ZHAN, Shenzhen (CN)**

(73) Assignee: **Autel Intelligent Technology Corp.,
Ltd., Shenzhen, GD (CN)**

(21) Appl. No.: **19/215,147**

(22) Filed: **May 21, 2025**

Related U.S. Application Data

(63) Continuation of application No. PCT/CN2023/
130107, filed on Nov. 7, 2023.

(30) **Foreign Application Priority Data**

Nov. 21, 2022 (CN) 202211458449.3

(57) **ABSTRACT**
Provided is a communication interaction method, including:
in a case where a vehicle identification number of a target
vehicle is acquired, selecting a plurality of electronic control
units and acquiring a plurality of communication parameters
in the target vehicle according to the vehicle identification
number, where the plurality of communication parameters
corresponds the plurality of electronic control units; creating
one or more communication links for the plurality of elec-
tronic control units according to the plurality of communi-
cation parameters, where at least one communication link
corresponds to at least one communication parameter; and
performing parallel communication command interaction
with the plurality of electronic control units corresponding
to the one or more communication links through the one or
more communication links.

S201

In the case where a vehicle identification number of a target
vehicle is acquired, selecting a plurality of electronic control units
in the target vehicle and acquiring a plurality of communication
parameters of the target vehicle according to the vehicle
identification number of the target vehicle

S202

Creating one or more communication links for one or more
electronic control units in the target vehicle according to the
plurality of communication parameters of the target vehicle

S203

Respectively performing parallel communication command
interaction with the one or more electronic control units
corresponding to the one or more communication links by means
of the one or more communication links

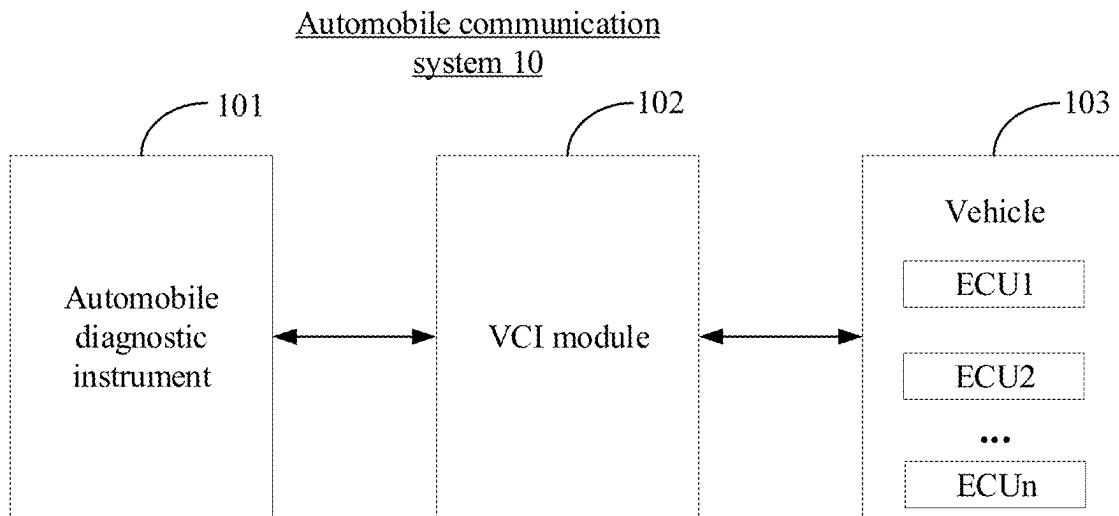
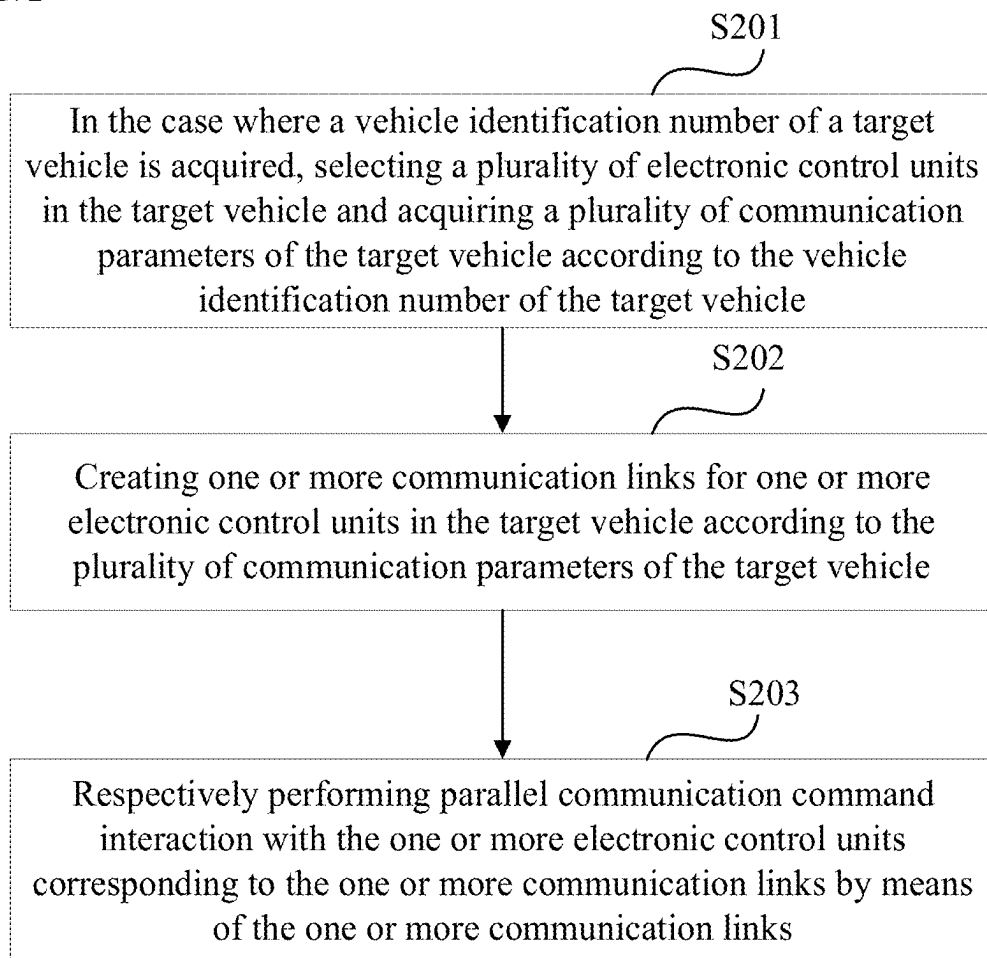


FIG. 1

FIG. 2



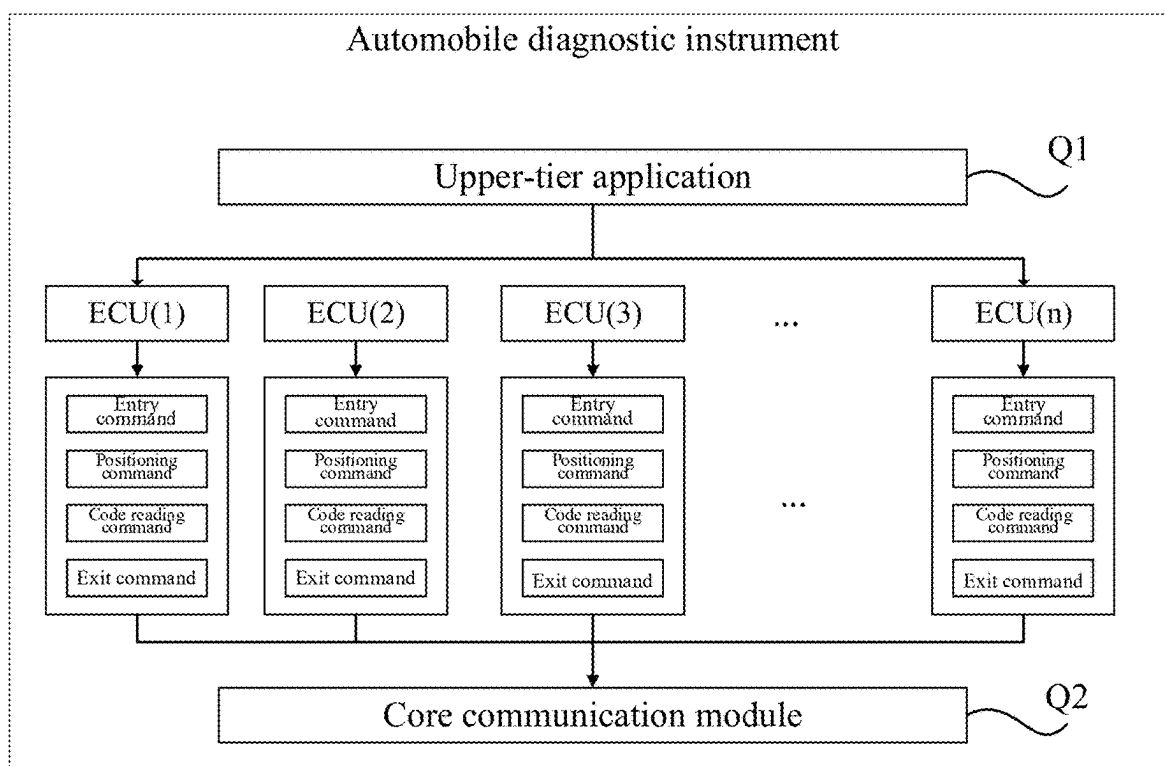


FIG. 3

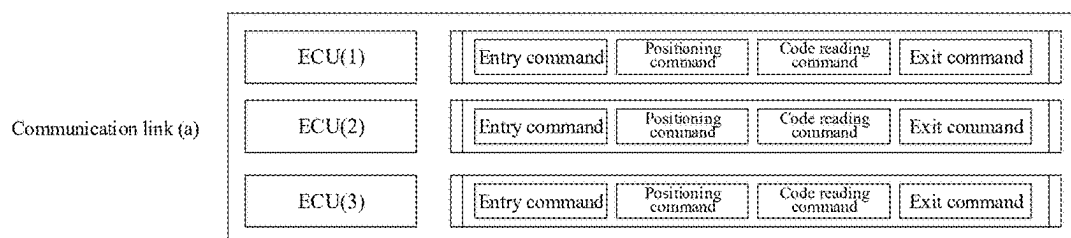


FIG. 4A

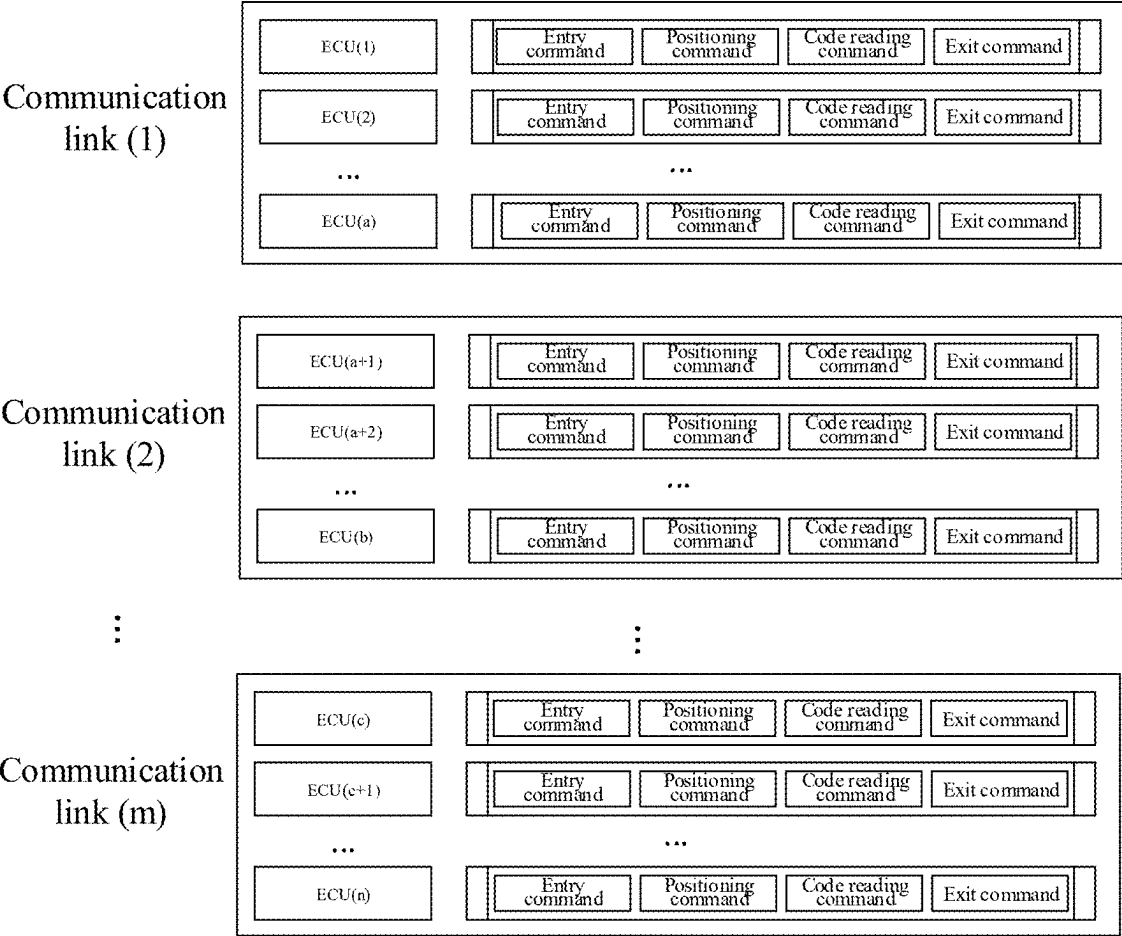


FIG. 4B

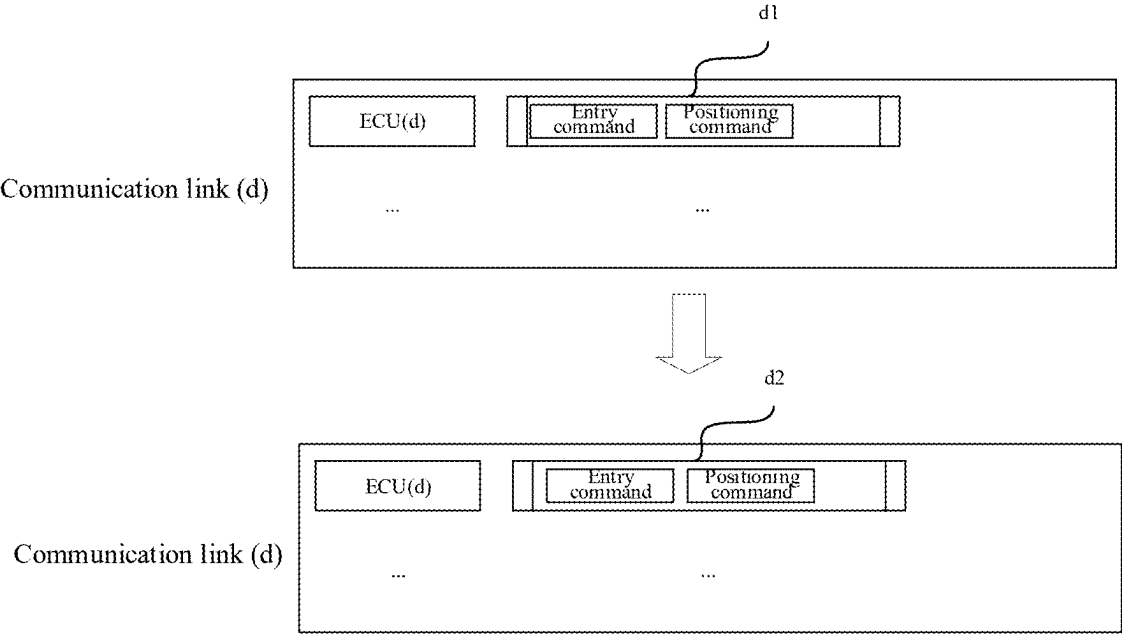


FIG. 4C

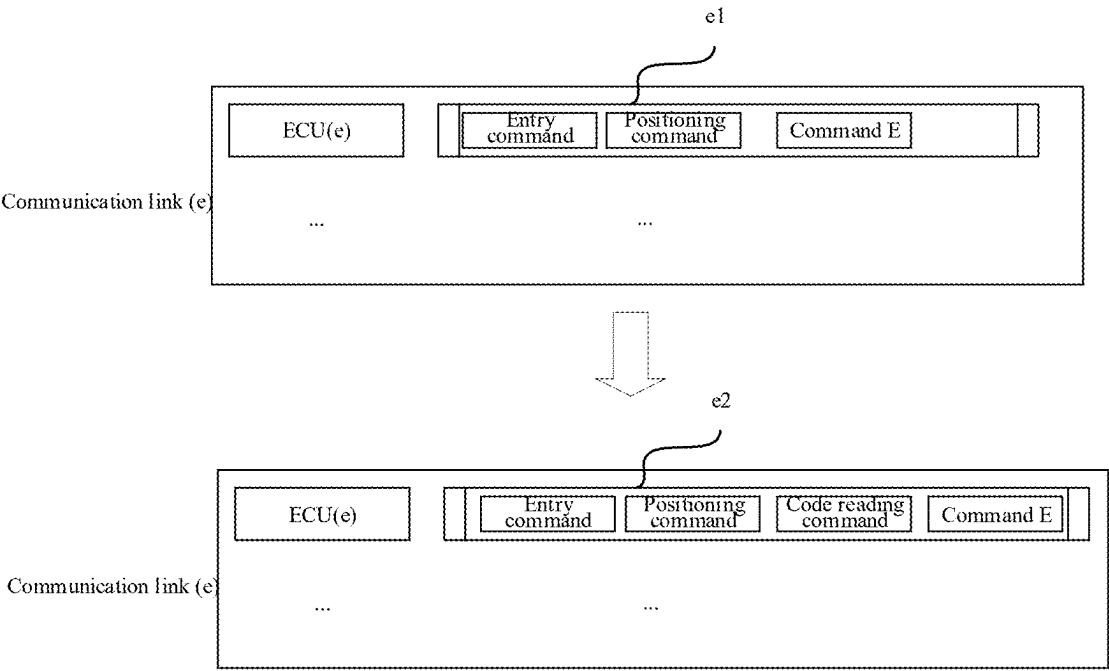


FIG. 4D

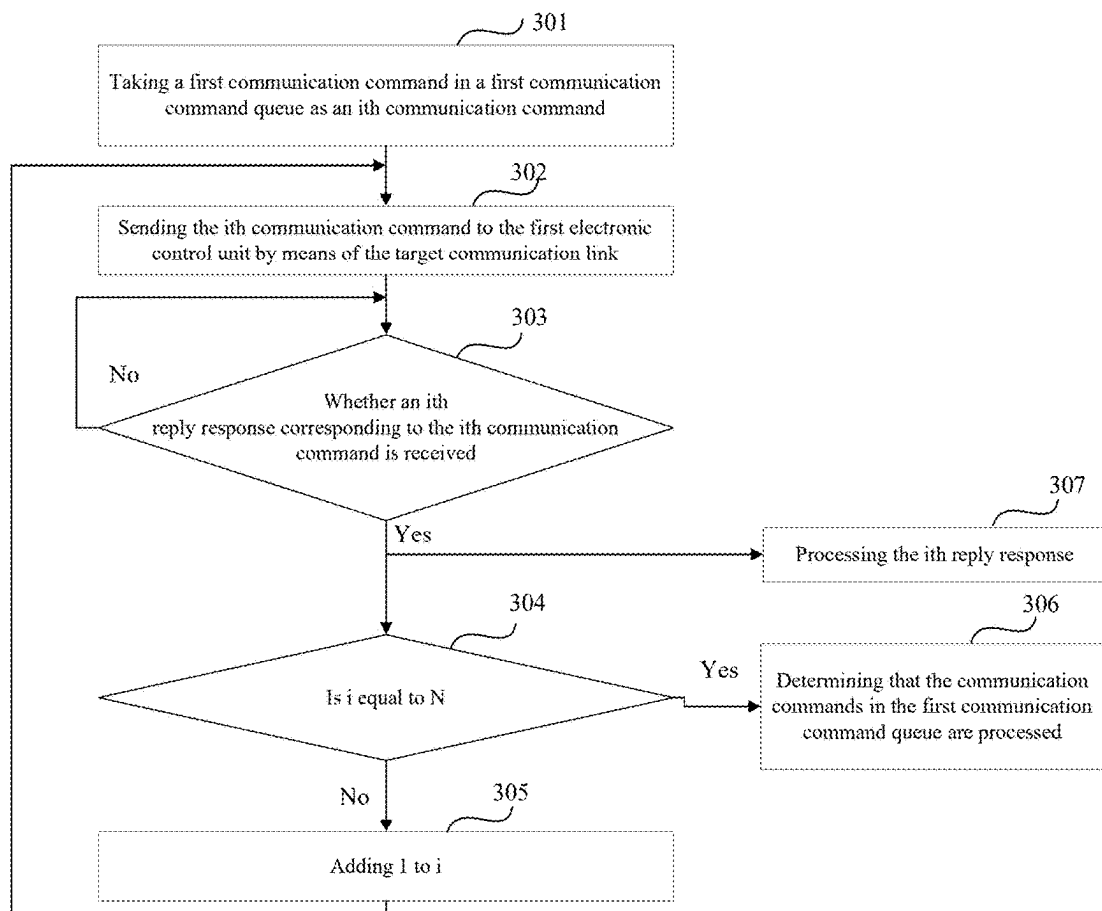


FIG. 5

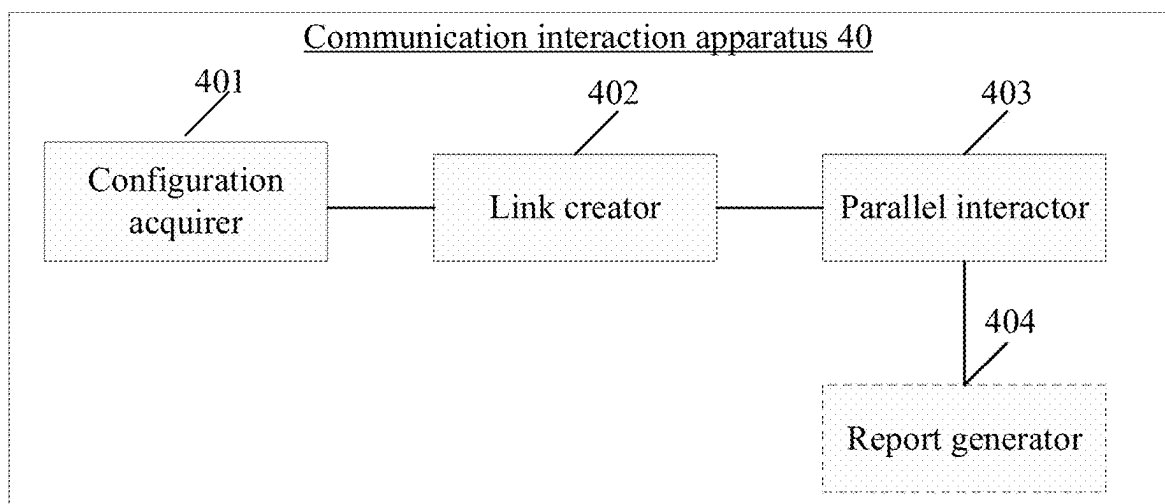


FIG. 6

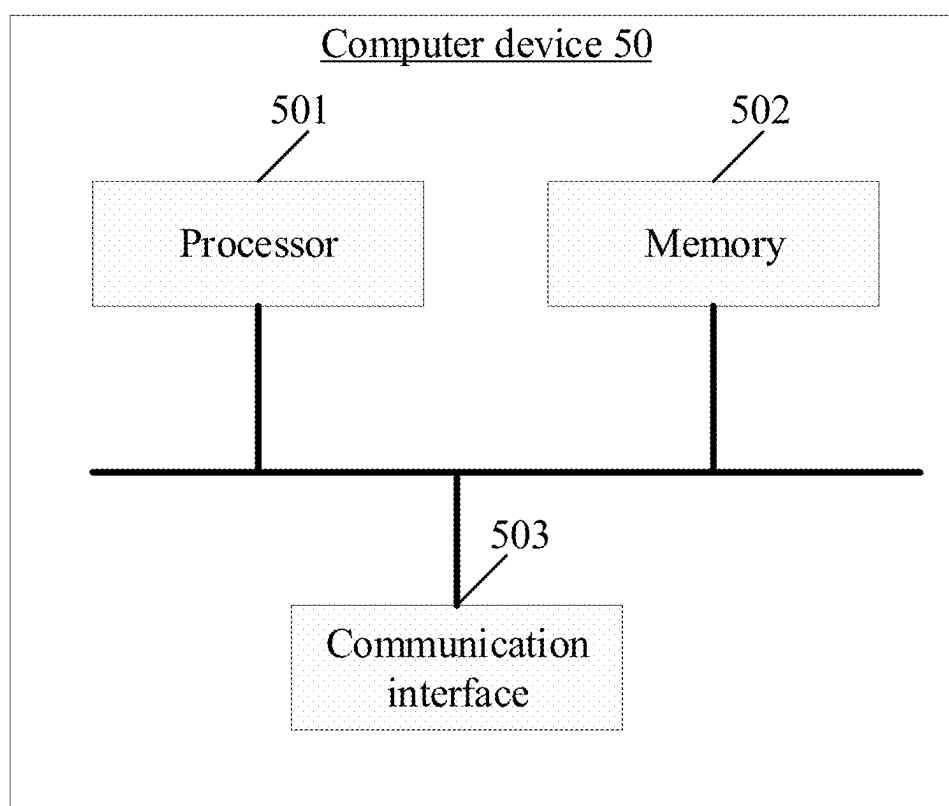


FIG. 7

COMMUNICATION INTERACTION METHOD, APPARATUS, DEVICE, AND STORAGE MEDIUM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of PCT Patent Application No. PCT/CN2023/130107, filed on Nov. 7, 2023, which claims priority to Chinese Patent Application No. 202211458449.3, filed on Nov. 21, 2022. The entire disclosures of the aforementioned applications are incorporated herein by reference for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of communications, and more particularly to a communication interaction method, an apparatus, a device, and a storage medium.

BACKGROUND

[0003] With the rapid development of the automobile industry, the progress of science and technology and the high demand of people, an automotive electronic technology has been continuously updated, and the number of electronic control units (ECUs) installed on an automobile increases to achieve different control functions.

[0004] The ECU has fault self-diagnosis and protection functions, and when a system is faulty, the ECU can automatically record a fault code in a random access memory (RAM) and turn on a fault light on an instrument panel for prompt. Fault information saved in the ECU may be read by means of an automobile diagnostic instrument, and the automobile diagnostic instrument may be connected to the automobile through an on board diagnostics (OBD) interface of the automobile, and communicates with the ECU in the automobile through the OBD interface, so as to acquire all the fault information in the automobile and parse the fault information to obtain information such as a current fault cause, a fault description and a fault state. At present, the automobile diagnostic instrument communicates with the ECU of the automobile based on a serial communication mode, namely, the automobile diagnostic instrument communicates with the ECUs one by one, and after finishing interaction with all communication commands between the automobile diagnostic instrument and a previous ECU one by one, the automobile diagnostic instrument performs interaction with communication commands between the automobile diagnostic instrument and a next ECU one by one, thereby consuming a relatively long time and being low in diagnosis efficiency.

SUMMARY

[0005] The present disclosure provides a communication interaction method, an apparatus, a device, and a storage medium to solve at least the technical problem that communication between an automobile diagnostic instrument and an ECU of an automobile consumes a relatively long time.

[0006] In a first aspect, provided is a communication interaction method, including:

[0007] in a case where a vehicle identification number of a target vehicle is acquired, selecting a plurality of electronic control units and acquiring a plurality of

communication parameters in the target vehicle according to the vehicle identification number, wherein the plurality of communication parameters corresponds to the plurality of electronic control units;

[0008] creating one or more communication links for the plurality of electronic control units according to the plurality of communication parameters, wherein at least one communication link corresponds to at least one communication parameter; and

[0009] performing parallel communication command interaction with the plurality of electronic control units corresponding to the one or more communication links through the one or more communication links.

[0010] In this aspect, after the vehicle identification number of the vehicle is acquired, firstly the plurality of electronic control units are selected and the communication parameters of the plurality of electronic control units in the vehicle are acquired according to the vehicle identification number of the vehicle; then the communication links are created for the plurality of electronic control units in the vehicle according to the communication parameter corresponding to each of the plurality of electronic control units, where one communication link corresponds to one communication parameter, so as to achieve classification of the electronic control units in the vehicle, and facilitate communication and transmission by allocating the electronic control units with the same communication parameters into one group; and finally the parallel communication command interaction with the plurality of electronic control units in the vehicle is performed through the created communication links, and the communication command interaction of the plurality of electronic control units in the vehicle is not dependent on each other, so that the duration of communication command interaction may be reduced, and the efficiency of communication with the electronic control units in the vehicle is improved.

[0011] In a second aspect, provided is a communication interaction apparatus, including:

[0012] an configuration acquirer configured to: in a case where a vehicle identification number of a target vehicle is acquired, select a plurality of electronic control units and acquiring a plurality of communication parameters in the target vehicle according to the vehicle identification number, where the plurality of communication parameters corresponds to the plurality of electronic control units;

[0013] a link creator configured to create one or more communication links for the plurality of electronic control units according to the plurality of communication parameters, where one communication link corresponds to one communication parameter; and

[0014] a parallel interactor configured to respectively perform parallel communication command interaction with the plurality electronic control units corresponding to the one or more communication links through the one or more communication links.

[0015] In a third aspect, provided is a computer device, including a memory and one or more processors, where the memory is connected to the one or more processors, the one or more processors are configured to execute one or more computer programs stored in the memory, and the one or more processors, when executing the one or more computer programs, cause the computer device to implement the above communication interaction method in the first aspect.

[0016] In a fourth aspect, provided is a non-transitory computer-readable storage medium storing a computer program, where the computer program includes a program instruction, and the above program instruction, when executed by a processor, causes the above processor to execute the above communication interaction method in the first aspect.

BRIEF DESCRIPTION OF DRAWINGS

[0017] FIG. 1 is a schematic diagram showing a system architecture of an automobile communication system according to some embodiments of the present disclosure;

[0018] FIG. 2 is a schematic flow diagram of a communication interaction method according to some embodiments of the present disclosure;

[0019] FIG. 3 is a schematic diagram showing interaction between an upper-tier application and a core communication module of an automobile diagnostic instrument according to some embodiments of the present disclosure;

[0020] FIG. 4A to FIG. 4D are schematic diagrams of communication command queues according to some embodiments of the present disclosure;

[0021] FIG. 5 is a schematic flow diagram showing transmitting and processing communication commands in one communication command queue over a communication link according to some embodiments of the present disclosure;

[0022] FIG. 6 is a schematic structural diagram of a communication interaction apparatus according to some embodiments of the present disclosure; and

[0023] FIG. 7 is a schematic structural diagram of a computer device according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0024] The technical solutions in embodiments of the present disclosure will be described below in conjunction with accompanying drawings in the embodiments of the present disclosure.

[0025] The technical solutions of the present disclosure may be applicable to an automobile communication scene, and specifically applicable to a scene in which an automobile diagnostic instrument communicates with a vehicle. In order to facilitate understanding, a communication architecture of the present disclosure will be firstly introduced. FIG. 1 is a schematic diagram of a system architecture of an automobile communication system according to some embodiments of the present disclosure. As shown in FIG. 1, the automobile communication system 10 includes an automobile diagnostic instrument 101, a vehicle communication interface (VCI) module 102 and a vehicle 103, where the automobile diagnostic instrument 101 is connected to the vehicle 103 by means of the VCI module 102, and where the automobile diagnostic instrument 101 may also be referred to as an automobile decoder used for communicating with an ECU in the vehicle 103 by means of a communication module such as a controller area network (CAN) communication module or a local interconnect network (LIN) communication module, and transmitting a fault code and engine state information of the vehicle 103, so as to achieve fault scanning detection of the vehicle 103; and the automobile diagnostic instrument 101 may also be used for vehicle ECU positioning, vehicle ECU data backup and data restoration, and the like for the vehicle 103. The VCI module

102 is an automobile communication connection tool used for being connected to an OBD interface of the vehicle 103 to construct communication connection between the automobile diagnostic instrument 101 and the vehicle 103. The VCI module 102 may be connected to the automobile diagnostic instrument 101 in the modes of Bluetooth, a universal serial bus (USB), WiFi, and the like. The VCI module 102 is connected to the OBD interface of the vehicle by means of a diagnostic main line. The vehicle 103 includes a plurality of ECUs, where the ECU consists of integrated circuits such as a microcontroller unit (MCU), an RAM, a read only memory (ROM) and an input/output interface, and different ECUs may be disposed at different mechanical apparatuses of the vehicle for detecting operation conditions of the different mechanical apparatuses and recording fault conditions of the different mechanical apparatuses. Specifically, the ECU may be disposed on the mechanical apparatus such as an engine and an anti-lock braking system of the vehicle 103.

[0026] Based on the system architecture shown in FIG. 1, the detection of fault information of the vehicle may be achieved. In some current solutions, the fault detection is performed by means of serial communication between the automobile diagnostic instrument 101 and the ECUs in the vehicle 103, namely, the automobile diagnostic instrument 101 performs communication command interaction with the ECUs one by one according to a precedence order, sends communication commands one by one and receives reply responses of the communication commands, sends the next communication command after receiving the reply response of the previous communication command, and performs communication interaction with the next ECU to obtain the fault information recorded in the next ECU after the communication interaction with the previous ECU is finished and the fault information recorded in the previous ECU is obtained. The number of the ECUs in the vehicle 103 is relatively large, so that a relatively long time is required to acquire all the fault information in the vehicle 103 in this mode, and the communication efficiency is low.

[0027] In view of this, the present disclosure provides a communication interaction method, including: grouping ECUs of a vehicle according to communication parameters of the ECUs in the vehicle, pooling the ECUs with the same communication parameters into one group, and creating communication links matching the communication parameters of the ECUs for transmitting communication commands of the ECUs with the same communication parameters; and in the same communication link, performing parallel transmission and interaction on the communication commands of the ECU corresponding to the communication link, so that the communication command interactions of the ECU do not interfere with each other, thereby reducing the communication interaction time, and improving the communication efficiency.

[0028] The technical solutions of the present disclosure are specifically introduced below.

[0029] With reference to FIG. 2, FIG. 2 is a schematic flow diagram of a communication interaction method according to some embodiments of the present disclosure, and the method may be applied to the above automobile diagnostic instrument 101. As shown in FIG. 2, the method includes the following steps:

[0030] S201, in the case where a vehicle identification number of a target vehicle is acquired, selecting a

plurality of electronic control units in the target vehicle and acquiring a plurality of communication parameters of the target vehicle according to the vehicle identification number of the target vehicle;

[0031] wherein the target vehicle is a vehicle which has established physical connection with the automobile diagnostic instrument by means of an VCI module. After establishing the physical connection with the target vehicle, the VCI module may turn on an ignition key of the target vehicle to start functional devices such as an engine of the target vehicle, so that communication connection is established between the automobile diagnostic instrument and the target vehicle, so as to acquire the vehicle identification number of the target vehicle.

[0032] The plurality of electronic control units in the target vehicle refer to a plurality of electronic control units with which communication is required for currently executing a certain task. For example, the automobile diagnostic instrument currently executes fault scanning for the target vehicle, objects of the fault scanning are all the electronic control units in the target vehicle, and then the plurality of electronic control units refer to all the electronic control units in the target vehicle. For another example, the automobile diagnostic instrument currently executes data backup for some of the electronic control units in the target vehicle, and then the plurality of electronic control units refer to the electronic control units which need data backup in the target vehicle. The number of the electronic control units depends on the task to be executed currently.

[0033] The plurality of communication parameters of the target vehicle include a communication parameter corresponding to each electronic control unit in the plurality of electronic control units in the target vehicle. The communication parameter corresponding to the electronic control unit refers to a parameter required to be adopted/followed for communication with the electronic control unit, and the communication parameter corresponding to the electronic control unit may include a communication protocol, a communication pin, a Baud rate, a time parameter, a communication command filling format, and the like adopted for communication with the electronic control unit.

[0034] In some embodiments, in the case where the vehicle identification number of the target vehicle is acquired, a model of the target vehicle may be obtained by parsing according to the vehicle identification number of the target vehicle, then a preset diagnosis database may be queried according to the model of the target vehicle to obtain all system sets and system configuration of the target vehicle, then a list of electronic control units may be acquired from all the system sets to obtain a plurality of electronic control units, and communication parameters of the plurality of electronic control units may be acquired from the system configuration.

[0035] S202, creating one or more communication links for the plurality of electronic control units in the target vehicle according to the plurality of communication parameters of the target vehicle;

[0036] wherein one communication link corresponds to one communication parameter, and different communication links correspond to different communication parameters. If the communication parameters of the plurality of electronic control units in the target vehicle are the same, the plurality of electronic control units

share the same communication link. The purpose of creating at least one communication link for the plurality of electronic control units in the target vehicle is to group the plurality of electronic control units in the target vehicle and construct an environment required for communication for the electronic control units in the same group; and

[0037] wherein a process of creating the communication link for the target vehicle may include filtering the ECU, and any one of target communication links in the at least one communication link is required to be set to allow to receive a communication message sent by the ECU corresponding to the target communication link, namely, one communication link is used for receiving the communication message sent by the ECU with a communication parameter the same as the communication parameter corresponding to the communication link.

[0038] In specific implementation, one electronic control unit may be selected from the plurality of electronic control units in the target vehicle as a target electronic control unit, and a target communication link matching a communication parameter of the target electronic control unit is created according to the communication parameter of the target electronic control unit; then electronic control units with communication parameters the same as the communication parameter of the target electronic control unit are determined according to communication parameters of other electronic control units in the target vehicle to form a first electronic control unit combination with the target electronic control unit; and finally the target communication link is set to allow to receive communication messages sent by the electronic control units in the first electronic control unit combination, so as to obtain one communication link. Communication links are created for electronic control units which have not been allocated into the electronic control unit combination in the target vehicle in the same mode until all the electronic control units in the target vehicle have corresponding communication links, and one or more communication links may be obtained. It should be understood that except that request addresses and reply addresses are different, other communication parameters are the same for each electronic control unit in the same electronic control unit combination.

[0039] It should be noted that the mode of creating the communication link in the present disclosure is basically the same as the principle of creating the communication link by serial communication between the automobile diagnostic instrument and the vehicle, with the difference that one communication link created when the automobile diagnostic instrument performs serial communication with the vehicle is allowed to receive a message of only one electronic control unit simultaneously; and in the present disclosure, since a plurality of communication links are created, and each communication link is allowed to receive messages of a plurality of electronic control units simultaneously, communication interaction with all the control units in the target vehicle is performed simultaneously.

[0040] S203, respectively performing parallel communication command interaction with the plurality electronic control units corresponding to the one or more communication links through the one or more communication links;

[0041] wherein performing parallel communication command interaction with the electronic control unit

corresponding to the communication link through the communication link refers to that for each created communication link, the communication commands of all the electronic control units corresponding to the communication link are transmitted in parallel through the communication link. The communication command of each electronic control unit may be acquired by an upper-tier application executing a certain task in the automobile diagnostic instrument, and after the upper-tier application of the automobile diagnostic instrument acquires the communication command of each electronic control unit, the communication command of each electronic control unit may be sent to a core communication module of the automobile diagnostic instrument. For example, the automobile diagnostic instrument currently performs a fault scanning task for a BMW automobile, and then the upper-tier application may refer to BMW diagnostic software in the automobile diagnostic instrument. The core communication module in the automobile diagnostic instrument may be, for example, a multi-channel communication module.

[0042] Taking the automobile diagnostic instrument performing a vehicle fault scanning service as an example, as shown in FIG. 3, after traversing to obtain all the ECUs (ECU1, ECU2, ECU3 . . . ECU_n) of the target vehicle, the upper-tier application Q1 of the automobile diagnostic instrument respectively acquires an entry command, a positioning command, a code reading command and an exit command of each ECU, and provides the commands to the core communication module Q2.

[0043] Each communication link transmits the communication commands of all the electronic control units corresponding to the communication link in parallel in the same mode. In one feasible mode, for one of the communication links, the parallel transmission of the communication command of each ECU may be completed in a mode of sequentially forming the command of each ECU corresponding to the communication link into a queue.

[0044] In some embodiments, the parallel transmission over the communication link may be achieved by the following steps A1 to A2.

[0045] A1, generating a communication command queue corresponding to each electronic control unit corresponding to a target communication link according to a communication command of each electronic control unit corresponding to the target communication link,

[0046] wherein the target communication link may be any one of the at least one communication link created in the above step 201.

[0047] One communication command queue includes all the communication commands of one electronic control unit corresponding to the target communication link.

[0048] In some embodiments, the communication command of each electronic control unit corresponding to the target communication link may be arranged into the communication command queue according to a command execution precedence order.

[0049] Still taking the automobile diagnostic instrument executing the vehicle fault scanning service as an example, assuming that communication parameters of ECU1, ECU2 and ECU3 in FIG. 3 are the same, and the communication link created for the ECU1, the ECU2 and the ECU3 is a

communication link a, a communication command queue corresponding to the generated communication link a may be as shown in FIG. 4A.

[0050] By performing the processing of step A1 on the electronic control unit corresponding to each communication link, the communication command queue corresponding to each communication link may be obtained.

[0051] Still taking the automobile diagnostic instrument executing the vehicle fault scanning service as an example, assuming that a total of m communication links are created according to the communication parameters of the ECUs in FIG. 3, the communication command queues corresponding to the m communication links may be as shown in FIG. 4B.

[0052] In some possible cases, the upper-tier application of the automobile diagnostic instrument may also newly add communication commands to the core communication module of the electronic control unit. After the above step A1, the method may further include: in the case where a first communication command of a first electronic control unit is acquired, inserting the first communication command into a first communication command queue,

[0053] wherein the first electronic control unit is any one of electronic control units corresponding to the target communication link, the first communication command is a newly added communication command, the first communication command queue is a communication command queue corresponding to the first electronic control unit, and the newly added communication command is generally not in the first communication command queue.

[0054] In some embodiments, the first communication command may be inserted at a position corresponding to a time sequence of the first communication command in the first communication command queue according to a time sequence precedence relationship between the first communication command and communication commands in the first communication command queue.

[0055] For example, with reference to FIG. 4C, assuming that the first electronic control unit is an ECU(d) in FIG. 4C, the communication command queue created for the ECU(d) according to the communication command of the ECU(d) is as shown in d1 in FIG. 4C; assuming that after the entry command of the ECU(d) is sent to the ECU(d) through a communication link (d), the upper-tier application sends a code reading command of the ECU(d) to the core communication module, since a time sequence of the code reading command is after a time sequence of a positioning command, the code reading command of the ECU(d) is inserted at the position after the positioning command, namely, at a tail of the communication command queue. After the code reading command is inserted at the tail of the communication command queue of ECU(d), the communication command queue of the ECU(d) is as shown in d2 in FIG. 4C.

[0056] For another example, with reference to FIG. 4D, assuming that the first electronic control unit is an ECU(e) in FIG. 4D, the communication command queue created for the ECU(e) according to the communication command of the ECU(e) is as shown in e1 in FIG. 4C; assuming that the upper-tier application sends a code reading command of the ECU(e) to the core communication module, a time sequence of the code reading command is after the time sequence of the positioning command and before a time sequence of a command E, the code reading command is inserted after the code reading command and before the command E. After the

code reading command is inserted into the communication command queue of the ECU(e), the communication command queue of the ECU(e) is as shown in e2 in FIG. 4C.

[0057] It should be understood that the newly added communication command may occur at any stage of parallel communication command interaction between the automobile diagnostic instrument and the target vehicle. The newly added communication command is inserted into the corresponding communication command queue, so that the newly added communication command can be processed, and the communication command interaction corresponding to the electronic control unit can be performed in an orderly manner.

[0058] A2, transmitting and processing the communication commands in the communication command queue corresponding to each electronic control unit corresponding to the target communication link in parallel through the target communication link;

[0059] wherein transmitting and processing the communication commands in the communication command queue corresponding to each electronic control unit corresponding to the target communication link in parallel refers to sequentially sending the communication commands in the command queue corresponding to each electronic control unit corresponding to the target communication link, and respectively receiving reply responses in the command queue corresponding to each electronic control unit corresponding to the target communication link. The sending of the communication commands and the receiving of the reply responses in each command queue do not interfere with each other.

[0060] Taking the target communication link being a communication link (a) of the above FIG. 4A as an example, a process of transmitting and processing the communication commands in the communication command queue corresponding to each electronic control unit corresponding to the target communication link in parallel is as follows: sequentially sending entry commands to the ECU1, the ECU2 and the ECU3, and when an entry reply response sent by the ECU1 is received, sending a positioning command to the ECU1; when an entry reply response sent by the ECU2 is received, sending a positioning command to the ECU2 . . . until exit commands of the ECU1 to the ECU3 have all been sent, and reply responses corresponding to the exit commands of the ECU1 to the ECU3 are received;

[0061] wherein for one communication command queue corresponding to the target communication link, reference may be made to FIG. 5 for a specific flow of transmitting and processing the communication command queue over the target communication link, and FIG. 5 is a schematic flow diagram of transmitting and processing communication commands in one communication command queue over a communication link according to some embodiments of the present disclosure. As shown in FIG. 5, the flow may include the following steps:

[0062] S301, taking a first communication command in a first communication command queue as an ith communication command;

[0063] wherein the first communication command queue is a communication command queue corresponding to the first electronic control unit, and the first

electronic control unit is any one of electronic control units corresponding to the target communication link;

[0064] S302, sending the ith communication command to the first electronic control unit through the target communication link;

[0065] S303, determining whether an ith reply response corresponding to the ith communication command is received;

[0066] in the case where the ith reply response corresponding to the ith communication command is received, performing step 304 and step 307; and in the case where the ith reply response corresponding to the ith communication command is not received, continuing to perform step 303;

[0067] S304, determining whether i is equal to N;

[0068] wherein N is a total number of communication commands contained in the first communication command queue;

[0069] when i is equal to N, indicating that a reply response corresponding to a last communication command in the first communication command queue has been received, performing step 306; and when i is not equal to N, indicating that there are communication commands which have not been sent in the first communication command queue, performing step 305;

[0070] S305, adding 1 to i, and returning to perform step 302;

[0071] S306, determining that the communication commands in the first communication command queue are processed; and

[0072] S307, processing the ith reply response.

[0073] In some embodiments, response data may be acquired from the ith reply response and analyzed to complete the processing of the ith reply response.

[0074] Communication command interaction with the electronic control unit may be completed by respectively processing each communication command queue corresponding to each communication link according to the above steps 301 to 307.

[0075] By sequentially processing the communication commands in the communication command queue, ordered interaction of the communication commands can be achieved.

[0076] In some possible cases, the above steps 301 to 306 may be performed by a first thread; and the above step 307 may be performed by a second thread, wherein the first thread and the second thread are two different threads. By completing the transceiving and processing of the communication commands by the different threads, the efficiency of communication command interaction can be improved.

[0077] In some possible cases, the processing of communication commands in different communication command queues may also be completed by different threads.

[0078] In some possible cases, after the above step 306, the flow may further include: generating a communication command processing result of the electronic control unit corresponding to the first communication command queue according to processing results of all the reply responses corresponding to the first communication command queue;

[0079] wherein the communication command processing result refers to a result of executing a certain task for the electronic control unit corresponding to the first communication command queue. Taking the automobile diagnostic instrument executing the vehicle fault

scanning service as an example, the communication command processing result of the electronic control unit corresponding to the first communication command queue refers to fault information of the electronic control unit corresponding to the first communication command queue.

[0080] After the communication commands in one communication command queue are processed, the communication command processing result of the electronic control unit is generated according to all the reply responses corresponding to the communication command queue, so as to achieve timely feedback on a communication command interaction result.

[0081] In the technical solution corresponding to the above FIG. 1, after the vehicle identification number of the vehicle is acquired, firstly the plurality of electronic control units in the vehicle are selected and the communication parameters of the plurality of electronic control units are acquired according to the vehicle identification number of the vehicle; then the communication links are created for the plurality of electronic control units in the vehicle according to the communication parameter corresponding to each of the plurality of electronic control units, wherein one communication link corresponds to one communication parameter, so as to achieve classification of the electronic control units in the vehicle, and facilitate communication and transmission by allocating the electronic control units with the same communication parameters into one group; and finally the parallel communication command interaction with the plurality of electronic control units in the vehicle is performed through the created communication links, and the communication command interaction of the plurality of electronic control units in the vehicle is not dependent on each other, so that the duration of communication command interaction may be reduced, and the efficiency of communication with the electronic control units in the vehicle is improved.

[0082] In some specific implementation scenes, the communication commands in the above solution of FIG. 1 may be communication commands used for acquiring fault information of the target vehicle. The above method of FIG. 1 may further include: after the communication commands in the communication command queue corresponding to each communication link are all transmitted and processed, generating a fault detection report corresponding to the target vehicle according to all the fault information acquired based on all the transmitted communication commands;

[0083] wherein the communication commands in the communication command queue corresponding to each communication link being all transmitted and processed refers to reply responses corresponding to all the communication commands being received and the processing of all the reply responses being completed. All the communication commands refer to communication commands in all the communication command queues. Taking FIG. 3A as an example, all the communication commands include communication commands of the ECU1 to the ECU_n.

[0084] In some embodiments, all the fault information of the target vehicle may be obtained according to the fault information contained in the reply responses corresponding to all the communication commands, so as to generate the fault detection report corresponding to the target vehicle.

[0085] After the communication command of each electronic control unit is transmitted and processed in parallel, the fault detection report of the vehicle is generated, thereby being conducive to the understanding of all the fault information of the vehicle by a user.

[0086] In some other possible cases, in the case where the electronic control units in the target vehicle supports broadcast instructions, for the above step 203 in FIG. 1, the broadcast instruction may also be sent to each electronic control unit through each communication link, and an instruction response sent by each electronic control unit in response to the broadcast instruction is received, so as to achieve parallel communication command interaction with each electronic control unit, and then acquire data of each electronic control unit in the target vehicle.

[0087] The method of the present disclosure is introduced above, and an apparatus of the present disclosure is introduced below.

[0088] FIG. 6 is a schematic structural diagram of a communication interaction apparatus according to some embodiments of the present disclosure, wherein the communication interaction apparatus may be the aforementioned automobile diagnostic instrument 101. As shown in FIG. 6, the communication interaction apparatus 40 includes:

[0089] a configuration acquirer 401 configured to: in the case where a vehicle identification number of a target vehicle is acquired, select a plurality of electronic control units and acquire a plurality of communication parameters in the target vehicle according to the vehicle identification number, where the plurality of communication parameters corresponds to the plurality of electronic control units;

[0090] a link creator 402 configured to create one or more communication links for the plurality of electronic control units according to the plurality of communication parameters, where one communication link corresponds to one communication parameter; and

[0091] a parallel interactor 403 configured to respectively perform parallel communication command interaction with the plurality of electronic control units corresponding to the one or more communication links through the one or more communication links.

[0092] In some embodiments, the above parallel interactor 403 is specifically configured to generate a communication command queue corresponding to each electronic control unit according to a communication command of each electronic control unit corresponding to a target communication link, where the target communication link is any one of the at least one communication link, and one communication command queue includes all the communication commands of one electronic control unit corresponding to the target communication link; and transmit and process the communication commands in the communication command queue corresponding to each electronic control unit in parallel through the target communication link.

[0093] In some embodiments, the above parallel interactor 403 is further configured to: in the case where a first communication command of a first electronic control unit is acquired, insert the first communication command into a first communication command queue, where the first electronic control unit is any one of electronic control units corresponding to the target communication link, the first communication command is a newly added communication

command, and the first communication command queue is a communication command queue corresponding to the first electronic control unit.

[0094] In some embodiments, the above parallel interactor **403** is specifically configured to insert the first communication command at a position corresponding to a time sequence of the first communication command in the first communication command queue according to a time sequence precedence relationship between the first communication command and communication commands in the first communication command queue.

[0095] In some embodiments, the above parallel interactor **403** is specifically configured to send an i th communication command in the first communication command queue to the first electronic control unit through the target communication link, where the first electronic control unit is an electronic control unit corresponding to the first communication command queue, the first electronic control unit is any one of electronic control units corresponding to the target communication link, $1 \leq i \leq N$, and N is a total number of the communication commands in the first communication command queue; and in the case where an i th reply response corresponding to the i th communication command sent by the first electronic control unit is received, process the i th reply response, and send an $(i+1)$ th communication command in the first communication command queue to the first electronic control unit through the target communication link.

[0096] In some embodiments, the above parallel interactor **403** is further configured to: after a reply response corresponding to a last communication command in the first communication command queue is received, generate a communication command processing result of the first electronic control unit according to processing results of all the reply responses corresponding to the first communication command queue.

[0097] In some embodiments, the above parallel interactor **403** is specifically configured to send the i th communication command in the first communication command queue to the first electronic control unit through the target communication link based on a first thread; and process the i th reply response based on a second thread, and send the $(i+1)$ th communication command in the first communication command queue to the first electronic control unit through the target communication link based on the first thread, where the second thread and the first thread are different threads.

[0098] In some embodiments, the communication command is used for acquiring fault information of the target vehicle; and the above apparatus further includes a report generator **404** configured to: after the communication commands in the communication command queue corresponding to each communication link are all transmitted and processed, generate a fault detection report corresponding to the target vehicle according to all the fault information acquired based on all the transmitted communication commands.

[0099] It should be noted that reference may be made to the description of the above method embodiments for the content which is not mentioned in some embodiments corresponding to FIG. 6, which will not be described in detail herein.

[0100] For the above apparatus, after the vehicle identification number of the vehicle is acquired, firstly the plurality of electronic control units and the communication param-

eters of the plurality of electronic control units in the vehicle are acquired according to the vehicle identification number of the vehicle; then the communication links are created for the plurality of electronic control units in the vehicle according to the communication parameter corresponding to each of the plurality of electronic control units, where one communication link corresponds to one communication parameter, so as to achieve classification of the electronic control units in the vehicle, and facilitate communication and transmission by allocating the electronic control units with the same communication parameters into one group; and finally the parallel communication command interaction with the plurality of electronic control units in the vehicle is performed through the created communication links, and the communication command interaction of the plurality of electronic control units in the vehicle is not dependent on each other, so that the duration of communication command interaction may be reduced, and the efficiency of communication with the electronic control units in the vehicle is improved.

[0101] FIG. 7 is a schematic structural diagram of a computer device according to some embodiments of the present disclosure, where the computer device may be the above automobile diagnostic instrument **101**. The computer device **50** includes a processor **501**, a memory **502** and a communication interface **503**, where the memory **502** and the communication interface **503** are connected to the processor **501**, for example, connected to the processor **501** by a bus.

[0102] The processor **501** is configured to support the computer device **50** to execute corresponding functions in the method in the above method embodiments. The processor **501** may be a central processing unit (CPU), a network processor (NP), a hardware chip or any combination thereof. The above hardware chip may be an application specific integrated circuit (ASIC), a programmable logic device (PLD) or a combination thereof. The above PLD may be a complex programmable logic device (CPLD), a field-programmable gate array (FPGA), a generic array logic (GAL) or any combination thereof.

[0103] The memory **502** is configured to store program codes, and the like. The memory **502** may include a volatile memory (VM), such as a random access memory (RAM); the memory **502** may also include a non-volatile memory (NVM), such as a read-only memory (ROM), a flash memory, a hard disk drive (HDD) or a solid-state drive (SSD); and the memory **502** may also include a combination of the memories of the above types.

[0104] The communication interface **503** is configured to send a communication command or receive a reply response corresponding to the communication command under the direction of the processor **501**. The communication interface **503** may be a hardware interface, such as a USB interface; and the communication interface **503** may also be a software interface, such as a Bluetooth interface and a WiFi interface, which is not limited herein.

[0105] The processor **501** may call the program codes to execute the following operation:

[0106] in the case where a vehicle identification number of a target vehicle is acquired, selecting a plurality of electronic control units and acquiring a plurality of communication parameters in the target vehicle according to the vehicle identification number, where the

plurality of communication parameters corresponds to the plurality of electronic control units;

[0107] creating at least one communication link for the plurality of electronic control units according to the plurality of communication parameters, where one communication link corresponds to one communication parameter; and

[0108] respectively performing parallel communication command interaction with the electronic control unit corresponding to each communication link through each communication link in the at least one communication link.

[0109] Some embodiments of the present disclosure further provide a computer-readable storage medium storing a computer program, where the computer program includes a program instruction, and the program instruction, when executed by a computer, causes the computer to execute the method according to the above embodiments.

[0110] It will be understood by those ordinarily skilled in the art that the implementation of all or part of the flow of the methods in the above embodiments may be completed by the computer program instructing the associated hardware, the program may be stored in the computer-readable storage medium, and the program, when executed, may include the flow of some embodiments of each method described above, wherein the storage medium may be a magnetic disk, an optical disk, a read-only memory (ROM) or a random access memory (RAM), and the like.

[0111] Although only the specific embodiments of the present disclosure have been described above, the scope of protection of the present disclosure is not limited thereto, any person skilled in the art would readily conceive of changes or substitutions within the technical scope disclosed in the present disclosure, and all these changes and substitutions should be included in the scope of protection of the present disclosure. Therefore, the scope of protection of the present disclosure should be defined by the scope of protection of the claims.

What is claimed is:

1. A communication interaction method, comprising:

in a case where a vehicle identification number of a target vehicle is acquired, selecting a plurality of electronic control units and acquiring a plurality of communication parameters in the target vehicle according to the vehicle identification number, wherein the plurality of communication parameters correspond to the plurality of electronic control units;

creating one or more communication links for the plurality of electronic control units according to the plurality of communication parameters, wherein at least one communication link corresponds to at least one communication parameter; and

performing parallel communication command interaction with the plurality of electronic control units corresponding to the one or more communication links through the one or more communication links.

2. The method according to claim 1, wherein performing the parallel communication command interaction with the plurality of electronic control units corresponding to the one or more communication links through the one or more communication links comprises:

generating one or more communication command queues corresponding to the plurality of electronic control units according to communication commands of the

plurality of electronic control units corresponding to a target communication link, wherein the target communication link is one of the one or more communication links, and one communication command queue comprises all of the communication commands of one electronic control unit corresponding to the target communication link; and

transmitting and processing the communication commands in the one or more communication command queues corresponding to the plurality of electronic control units in parallel through the target communication link.

3. The method according to claim 2, wherein after generating the one or more communication command queues corresponding to the plurality of electronic control units according to the communication commands of the electronic control unit corresponding to the target communication link, the method further comprises:

in a case where a first communication command of a first electronic control unit is acquired, inserting the first communication command into a first communication command queue corresponding to the first electronic control unit, wherein the first electronic control unit is one of the plurality of electronic control units corresponding to the target communication link, the first communication command is a newly added communication command.

4. The method according to claim 3, wherein inserting the first communication command into the first communication command queue corresponding to the first electronic control unit comprises:

inserting the first communication command at a position in the first communication command queue according to a time sequence precedence relationship between the first communication command and communication commands in the first communication command queue.

5. The method according to claim 2, wherein transmitting and processing the communication commands in the one or more communication command queues corresponding to the plurality of electronic control units in parallel through the target communication link, comprises:

sending an i th communication command in a first communication command queue to a first electronic control unit through the target communication link, wherein the first electronic control unit corresponds to the first communication command queue, and is one of the plurality of electronic control units corresponding to the target communication link, wherein $1 \leq i \leq N$, and N is a total number of communication commands in the first communication command queue; and

in a case where an i th reply response corresponding to the i th communication command sent by the first electronic control unit is received, processing the i th reply response, and sending an $(i+1)$ th communication command in the first communication command queue to the first electronic control unit through the target communication link.

6. The method according to claim 5, further comprising:

after a reply response corresponding to a last communication command in the first communication command queue is received, generating a communication command processing result of the first electronic control

unit according to processing results of all reply responses corresponding to the first communication command queue.

7. The method according to claim 5, wherein sending the *i*th communication command in the first communication command queue to the first electronic control unit through the target communication link comprises:

sending the *i*th communication command in the first communication command queue to the first electronic control unit through the target communication link based on a first thread; and wherein the processing the *i*th reply response, and sending the (*i*+1)th communication command in the first communication command queue to the first electronic control unit through the target communication link, comprises:

processing the *i*th reply response based on a second thread, and sending the (*i*+1)th communication command in the first communication command queue to the first electronic control unit through the target communication link based on the first thread, wherein the second thread and the first thread are different threads.

8. The method according to claim 2, further comprising: after the communication commands in the one or more communication command queue corresponding to the one or more communication links are all transmitted and processed, generating a fault detection report corresponding to the target vehicle according to all fault information acquired based on all of the transmitted communication commands, wherein the communication commands are configured for acquiring the fault information of the target vehicle.

9. A computer device, comprising a memory, a processor, and a communication interface, wherein the communication interface and the memory are connected to the processor, the processor is configured to execute one or more computer programs stored in the memory, and the processor, when executing the one or more computer programs, causes the computer device to perform acts comprising:

in a case where a vehicle identification number of a target vehicle is acquired, selecting a plurality of electronic control units and acquiring a plurality of communication parameters in the target vehicle according to the vehicle identification number, wherein the plurality of communication parameters corresponds to the plurality of electronic control units;

creating one or more communication links for the plurality of electronic control units according to the plurality of communication parameters, wherein at least one communication link corresponds to at least one communication parameter; and

performing parallel communication command interaction with the plurality of electronic control units corresponding to the one or more communication links through the one or more communication links.

10. The computer device according to claim 9, wherein performing the parallel communication command interaction with the plurality of electronic control units corresponding to the one or more communication links through the one or more communication links comprises:

generating one or more communication command queues corresponding to the plurality of electronic control units according to communication commands of the plurality of electronic control units corresponding to a target communication link, wherein the target commu-

nication link is one of the one or more communication links, and one communication command queues comprises all of the communication commands of one electronic control unit corresponding to the target communication link; and

transmitting and processing the communication commands in the one or more communication command queues corresponding to the plurality of electronic control units in parallel through the target communication link.

11. The computer device according to claim 10, wherein after generating the one or more communication command queues corresponding to the plurality of electronic control units according to the communication commands of the electronic control unit corresponding to the target communication link, the processor causes the computer device to perform acts further comprising:

in a case where a first communication command of a first electronic control unit is acquired, inserting the first communication command into a first communication command queue corresponding to the first electronic control unit, wherein the first electronic control unit is one of the plurality of electronic control units corresponding to the target communication link, the first communication command is a newly added communication command.

12. The computer device according to claim 11, wherein inserting the first communication command into the first communication command queue corresponding to the first electronic control unit comprises:

inserting the first communication command at a position in the first communication command queue according to a time sequence precedence relationship between the first communication command and communication commands in the first communication command queue.

13. The computer device according to claim 10, wherein transmitting and processing the communication commands in the one or more communication command queues corresponding to the plurality of electronic control units in parallel through the target communication link comprises:

sending an *i*th communication command in a first communication command queue to a first electronic control unit through the target communication link, wherein the first electronic control unit corresponds to the first communication command queue, and is one of the plurality of electronic control units corresponding to the target communication link, wherein $1 \leq i \leq N$, and *N* is a total number of communication commands in the first communication command queue; and

in a case where an *i*th reply response corresponding to the *i*th communication command sent by the first electronic control unit is received, processing the *i*th reply response, and sending an (*i*+1)th communication command in the first communication command queue to the first electronic control unit through the target communication link.

14. The computer device according to claim 13, further comprising:

after a reply response corresponding to a last communication command in the first communication command queue is received, generating a communication command processing result of the first electronic control

unit according to processing results of all reply responses corresponding to the first communication command queue.

15. The computer device according to claim **13**, wherein sending the *i*th communication command in the first communication command queue to the first electronic control unit through the target communication link comprises:

sending the *i*th communication command in the first communication command queue to the first electronic control unit through the target communication link based on a first thread; and

wherein the processing the *i*th reply response, and sending the (*i*+1)th communication command in the first communication command queue to the first electronic control unit through the target communication link, comprises:

processing the *i*th reply response based on a second thread, and sending the (*i*+1)th communication command in the first communication command queue to the first electronic control unit through the target communication link based on the first thread, wherein the second thread and the first thread are different threads.

16. The computer device according to claim **10**, further comprising:

after the communication commands in the one or more communication command queue corresponding to the one or more communication links are all transmitted and processed, generating a fault detection report corresponding to the target vehicle according to all fault information acquired based on all of the transmitted communication commands, wherein the communication commands are configured for acquiring the fault information of the target vehicle.

17. A non-transitory computer-readable storage medium storing a computer program, wherein the computer program comprises a program instruction, and the program instruction, when executed by a processor, causes the processor to perform acts comprising:

in a case where a vehicle identification number of a target vehicle is acquired, selecting a plurality of electronic control units and acquiring a plurality of communication parameters in the target vehicle according to the vehicle identification number, wherein the plurality of communication parameters corresponds to the plurality of electronic control units;

creating one or more communication links for the plurality of electronic control units according to the plurality of communication parameters, wherein at least one communication link corresponds to at least one communication parameter; and

performing parallel communication command interaction with the plurality of electronic control units corresponding to the one or more communication links through the one or more communication links.

18. The storage medium according to claim **17**, wherein performing the parallel communication command interaction with the plurality of electronic control units corresponding to the one or more communication links through the one or more communication links comprises:

generating one or more communication command queues corresponding to the plurality of electronic control units according to communication commands of the plurality of electronic control units corresponding to a target communication link, wherein the target communication link is one of the one or more communication links, and one communication command queues comprises all of the communication commands of one electronic control unit corresponding to the target communication link; and

transmitting and processing the communication commands in the one or more communication command queues corresponding to the plurality of electronic control units in parallel through the target communication link.

19. The storage medium according to claim **18**, wherein after the generating one or more communication command queues corresponding to the plurality of electronic control units according to the communication commands of the electronic control unit corresponding to the target communication link, the program instruction causes the processor to perform acts further comprising:

in a case where a first communication command of a first electronic control unit is acquired, inserting the first communication command into a first communication command queue corresponding to the first electronic control unit, wherein the first electronic control unit is one of the plurality of electronic control units corresponding to the target communication link, the first communication command is a newly added communication command.

20. The storage medium according to claim **19**, wherein inserting the first communication command into the first communication command queue corresponding to the first electronic control unit comprises:

inserting the first communication command at a position in the first communication command queue according to a time sequence precedence relationship between the first communication command and communication commands in the first communication command queue.

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